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CS435 - Cloud Computing.

LEARNING MATERIALS

1 - 288

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1 - 115.

- Essential Characteristics according to NIST definition:
  - Resource pooling
  - Rapid elasticity
  - On-demand self-service
  - Broad network access
  - Measured service

- Important Benefits of Cluster Computing:
  - Scalability
  - High availability and fault tolerance
  - Use of commodity computers (Widely available/affordable)

- Service Models according to NIST definition:
  - Infrastructure as a Service (IaaS)
  - Platform as a Service (PaaS)
  - Software as a Service (SaaS)

- Cloud Deployment Models according to NIST definition:
  - Private cloud
  - Community cloud
  - Public cloud
  - Hybrid cloud

- Servers are available in a variety of sizes and types:
  - Web server
  - Domain Name Server (DNS)
  - Database server

- Two common types of switched networks are:
  - Packet-Switched Network
  - Circuit-Switched Network

- TCP/IP Protocol Stack Layers:
  - Application

- Transport
- Network
- Data-link
- Physical

- Main protocols of Transport Layer are:
  - Transmission Control Protocol (TCP)
  - User Datagram Protocol (UDP)

- IP Addressing
  - IPv4 is a 32 bit universally unique address while IPv6 is the 128 bit universally unique address.
    - Total IPv4 addresses = 2^{32}
    - Total IPv6 addresses = 2^{128}

- Classification of IPv4 addresses:

	Bits For Network ID	**Total Networks**	**Network ID Starts With**	**First Byte**
Class A	8	2^7	'0' binary	0 to 127
Class B	16	2^{14}	'10' binary	128 to 191
Class C	24	2^{21}	'110' binary	192 to 223
Class D	Used for multicasting and No prefix or Network ID			224 to 239
Class E	Reserved for future use			240 to 255

- There are three categories of IPv6 addresses:
  - Unicast address
    - A unique address that identifies only one device or network. It's like a personal phone number that only rings on your phone.
  - Multicast address
    - A special address that sends data to multiple devices or networks at the same time. It's like a group phone number that rings on multiple phones at once.
  - Anycast address
    - A single address that can be used by multiple devices or networks, but data is only sent to the closest one. It's like a shared phone number that automatically connects you to the nearest available phone.

- Standard Ethernet types
  - Bridged
  - Switched
  - Full duplex switches

- Wired LAN vs. Wireless LAN

Wired LAN	Wireless LAN
Medium: Wires	Medium: Air
Physical connection to network	No physical connection to network
Broadcasting and multicasting possible when required	All devices are broadcasting
Hosts are connected through link layer switch	No link layer switch exists
Connection to other networks through router	Connected to other networks through access point (a device that connects a wireless and wired network)

- Advantages of Switch:
  - Connecting heterogeneous devices
  - Collision elimination

- Types of routing
  - Unicast routing
  - Multicast routing

  - Connectionless routing
  - Connection oriented routing

- Virtualization implementation levels:
  - Instruction Set Architecture (ISA) level
  - Hardware Abstraction level
  - Operating System Level
  - Library support level
  - Application level

- Classes of VM architectures
  - Hypervisor Architecture
  - Full-virtualization Architecture
  - Para-virtualization Architecture

- Reasons of virtualization:
  - Sharing of resources
  - Isolation of users of shared resource
  - Dynamic provisioning of virtual resources is easier than physical resources
  - Aggregation of smaller resources into a single big virtual resource
  - Easier management of virtual resources.

- A VM can be in any of the following states:
  - Powered-off
  - Suspended
  - Paused
  - Powered-on

- The following options are available for VM migration:
  - Cold migration
  - Warm migration
  - Live migration

- A VM is made of two basic components:
  - VM state (The processor and RAM contents)
  - Virtual hard disk (Residing on network storage or on host's hard disk)

- The data center owner has three major considerations:
  - Assuring Performance and QoS
  - Increase resource utilization
  - Saving costs

- Types of VPN:
  - Site-to-site VPN
  - Remote-access VPN:

- Business drivers lure the organizations to start using Cloud.
  - IT Capacity Planning
    - Lead Strategy
    - Lag Strategy
    - Match Strategy
  - Organizational Agility (Rapidly meet customer demands and expectations & Maintain a continuous competitive advantage)

- Cost Reduction

- Types of scaling:
  - Horizontal scaling
  - Vertical scaling

- Advantages of network storage (particularly of SAN) are:
  - Compatibility with common file systems and operating systems
  - Better performance than file server
  - Data reliability and reconstruction through replication
  - Best choice for backups

- Cloud based Database solutions:
  - Advantages
    - Cost effective scalability as per use
    - High availability of data due to replication of database
    - Reduced administration of database provided as service or as part of PaaS.
  - Disadvantages
    - Due to Internet based access, the Cloud based database is not as fast as a locally installed database
    - The user may not trust the cloud provider regarding sensitive data

- General Characteristics of Cloud according to NIST:
  - Network dependence
  - Workload locations are hidden from consumer
  - Security risks due to multi-tenency
  - IT skills required

- Common Characteristics of Multi-tenant Applications:
  - Scalability in terms of number of tenants
  - Usage isolation
  - Databases, tables and/or schema isolation for each user
  - Backup and restore is separate for each tenant
  - Data security

- Privacy issues of Cloud Computing:
  - Lack of user control

- Lack of training and expertise
- Possibility of secondary unauthorized use of consumer data
- Legal compliance

- Security issues of Cloud Computing:
  - Insufficient monitoring and audit
  - Lack of security policies
  - Unauthorized access
  - No trust of data deletion
  - Backup vulnerabilities

- Types of Cloud usage monitors:
  - Monitoring Agent
  - Resource Agent
  - Polling Agent

- The agreement has two parts:
  - Service Agreement
  - Service Level Agreement (SLA)

- Major goals of a Cloud platform can be:
  - Virtualization
  - Scalability
  - Efficiency
  - Reliability

- Configurations [Failover System]
  - Active-Active
  - Active-Passive

- Cluster Computing v/s Grid Computing

Cluster Computing	**Grid Computing**
Computers are located close to each other.	Computers may be located at a huge distance from one another.
Computers are connected by a high speed local area network bus.	Computers are connected using a low speed bus or the internet.
Computers are connected in a centralized network topology.	Computers are connected in a distributed or de-centralized network topology.

Whole system has a centralized resource manager.	Every node manages it’s resources independently.
Whole system functions as a single system.	Each node behaves independently.